

GPD: Physiological Signal-Guided Progressive Cross-Modal Distillation for Non-Contact Deception Detection

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Abstract

Non-contact deception detection remains challenging because visual and auditory deception cues often lack stable cross-subject patterns. To improve non-contact deception detection, we explore cross-modal knowledge distillation (CMKD) from galvanic skin response (GSR), which provides more reliable physiological evidence of deception, to guide visual and auditory representation learning. To enable this study, we introduce MuDD, a large-scale multimodal deception detection dataset collected under the Guilty Knowledge Test (GKT) paradigm, with recordings from 130 participants spanning 690 minutes across six modalities. Building on MuDD, we propose GSR-guided Progressive Distillation (GPD), a cross-modal distillation framework that models knowledge transfer as a dynamic process. Unlike existing CMKD methods that typically rely on predefined distillation schemes, GPD adaptively regulates both the transfer pathway and the transferred knowledge as the modality gap evolves. Specifically, GPD uses gap-aware dynamic routing to select distillation configurations based on the evolving

representational gap between teacher and student, and progressively adjusts the relative importance of feature-level and logit-level knowledge during training for stable cross-modal transfer. Extensive experiments and visualizations show that GPD outperforms existing methods and achieves state-of-the-art performance on both deception detection and concealed-digit identification.

CCS Concepts

• **Computing methodologies** → **Neural networks**; *Learning latent representations*.

Keywords

non-contact deception detection, multimodal learning, cross-modal distillation, galvanic skin response, representation learning

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1 Introduction

Deception is a fundamental yet elusive aspect of human social behavior, with important implications for security screening, forensic

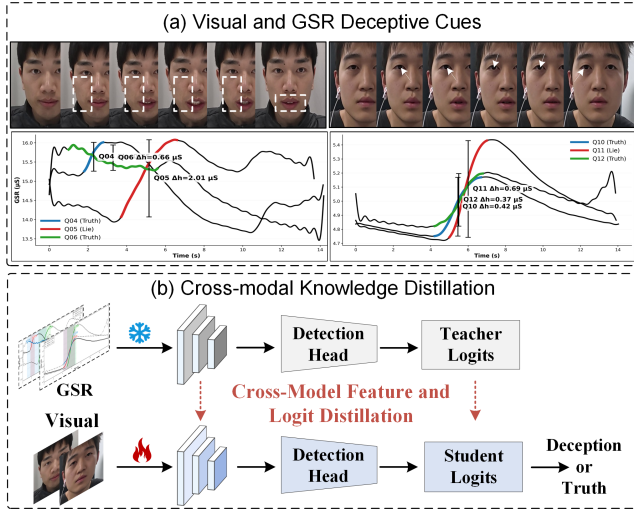


Figure 1: Motivation of this work. (a) Visual deception cues are often subtle and subject-dependent, making it difficult to identify stable cross-subject patterns, whereas GSR exhibits more consistent deception-related responses across individuals. (b) Illustration of the cross-modal knowledge distillation paradigm, where GSR serves as the teacher modality to transfer feature-level and logit-level knowledge to non-contact modalities through a teacher-student framework.

investigation, and interpersonal communication. Despite its practical significance, unaided human ability to detect deception remains only slightly above chance level [4, 8], highlighting the inherent difficulty of relying on subjective judgment alone. This limitation has driven the development of more objective approaches for deception assessment. Early studies primarily relied on contact-based polygraph methods, which infer deceptive behavior by measuring physiological responses associated with autonomic nervous system activity, such as respiration, heart rate (HR), and galvanic skin response (GSR), under standardized testing protocols [3, 40].

Although contact-based polygraph methods have demonstrated relatively high reliability, their practical applicability remains limited by the need for specialized equipment and controlled examination procedures. To address these limitations, recent research has increasingly turned to non-contact deception detection, which seeks to infer deceptive behavior from overt and non-invasive signals, such as facial expressions and linguistic cues [9, 10, 22]. However, the performance of existing non-contact methods on public benchmarks remains modest and inconsistent across datasets [5, 35, 45], with reported performance typically ranging from 63% to 73%¹. Although these results are not directly comparable due to differences in datasets and evaluation metrics, their substantial variation and overall moderate performance nevertheless reflect the difficulty of non-contact deception detection.

One important reason is that deception cues in audio-visual modalities are often subtle and highly subject-dependent, making it difficult to learn stable and generalizable patterns across individuals

[48]. As illustrated in Fig. 1(a), two deceptive cases from different subjects exhibit markedly different visual characteristics. The subject on the left shows relatively pronounced facial changes under deception, whereas the subject on the right exhibits only weak visual variation, without clearly salient facial cues. This variability suggests that visual deception cues are often neither robust nor consistently shared across subjects. In contrast, the corresponding GSR signals exhibit a clearer cross-subject regularity, where the **red** curve denotes the GSR response to a deceptive question, while the **green** and **blue** curves correspond to truthful responses to the immediately preceding and following questions. Although the absolute signal magnitude and waveform differ across subjects, the deceptive response consistently exhibits larger fluctuations than its adjacent truthful counterparts.

This observation raises a natural question: *can the more stable discriminative patterns in GSR be exploited to improve representation learning in non-contact modalities?* A natural paradigm for addressing this question is cross-modal knowledge distillation (CMKD). As illustrated in Fig. 1(b), CMKD provides a general teacher-student framework in which GSR can serve as the teacher modality, while the target non-contact modality is treated as the student. Through distillation objectives, the student is encouraged to absorb discriminative knowledge from GSR, thereby potentially improving deception detection performance when only the non-contact modality is available at test time [42]. Although CMKD has demonstrated effectiveness in other domains [1, 26, 41, 47], its application to non-contact deception detection faces two challenges. First, there is a lack of a large-scale dataset for studying how GSR can be leveraged to improve non-contact deception detection. Second, the large heterogeneity between GSR and non-contact modalities makes effective cross-modal transfer difficult. Existing CMKD methods have explored various ways to alleviate this heterogeneity, either by making teacher knowledge more transferable or by improving the transfer process [2, 18, 18, 23, 24, 31, 41, 43, 44, 46]. Despite their effectiveness, most of these methods do not explicitly model cross-modal distillation as an evolving process during training, which may limit effective knowledge transfer under large modality discrepancy. To address these issues, we first construct a large-scale Multimodal Deception Detection Dataset (MuDD), containing 130 subjects and 690 minutes of recordings. In addition to the task-required video, audio, and GSR modalities, the dataset also provides Photoplethysmography (PPG), heart rate, and personality trait information, supporting a wide range of deception-related research. Based on MuDD, we propose GSR-guided Progressive Distillation (GPD), a cross-modal distillation framework for non-contact deception detection. Unlike existing CMKD methods that typically rely on predefined distillation schemes, GPD models cross-modal distillation as a dynamic adaptation process. Specifically, it jointly incorporates two types of distillation knowledge and four candidate transfer configurations. According to the evolving representational gap between teacher and student, GPD uses gap-aware dynamic routing to adaptively select suitable transfer configurations, and employs progressive weighting to regulate the relative importance of feature-level and logit-level knowledge over training. Through this joint adaptation of transfer pathway and transferred knowledge, GPD facilitates cross-modal knowledge transfer.

The main contributions of this work are summarized as follows:

¹Performance denotes either accuracy or recall.

Table 1: Comparison of MuDD with existing deception-related datasets. Abbreviations: Subs. = subjects, Dur. = duration, V = video, A = audio, T = text, Pers. = personality traits, Bin. = binary label, and T-10cls = trial-level 10-class label.

Dataset	Subs.	Dur.	Setting	Modalities	Labels
<i>Without physiological signals</i>					
CSC Deceptive Speech [17]	32	7920 min	Interview	A	Bin.
CXD Corpus [21]	340	7200 min	False interview	A+T	Bin.
Real-life Trial [33]	56	< 60 min	Courtroom	V+T	Bin.
MU3D [27]	80	< 200 min	Social interview	V+T	Bin.
Truth-Tellers [29]	60	< 480 min	Simulated interview	V	Bin.
Box of Lies [35]	26	144 min	Bluffing game	V+T	Bin.
DOLOS [13]	213	≈ 530 min	Daily deception game	V+A	Bin.
MDPE [5]	193	6209 min	Simulated interview	V+A+T+Pers.	Bin.
<i>With physiological signals</i>					
GSR Deception Dataset [30]	20	< 320 min	Bluffing game	GSR	Bin.
Bag of Lies [15]	35	≈ 110 min	Daily lying	V+A+EEG+Gaze	Bin.
SEUMLD [45]	76	≈ 366 min	Simulated interview	V+A+ECG	Bin.
CogniModal-D [19]	≈ 110	–	Scenario task	EEG+ECG+Gaze+GSR+A+T	Bin.
MuDD (ours)	130	690 min	GKT	V+A+GSR+PPG+HR+Pers.	Bin.+T-10cls

- We construct MuDD, a large-scale multimodal deception detection dataset collected under the classical Guilty Knowledge Test (GKT) paradigm, providing a benchmark for a wide range of automatic deception detection studies.
- We propose GPD, a GSR-guided progressive CMKD framework for non-contact deception detection. The proposed framework dynamically models the overall process of CMKD through gap-aware routing and progressive weighting.
- Extensive experiments and visualization analyses show that, compared with existing methods, GPD enables the student to learn teacher knowledge effectively, resulting in more teacher-aligned decision behavior and better performance.

2 RELATED WORK

2.1 Multimodal Deception Detection Datasets

Existing deception detection datasets span a wide range of settings, including courtroom videos [33], simulated interviews [5, 17, 21, 29], social narratives [27], and game-based interactions [13, 35], but most provide only audio-visual or textual modalities without physiological signals (Table 1). Some datasets have introduced physiological signals such as EEG and ECG [15, 45], yet they still lack GSR, a widely used and relatively reliable indicator in deception research [38]. The GSR Deception Dataset [30] provides GSR but is limited to single-modality data, while CogniModal-D [19] incorporates both GSR and overt signals but is not collected under classical polygraph-style paradigms (e.g., CQT or GKT) designed to induce deception-related autonomic arousal. In contrast, MuDD synchronously acquires GSR and non-contact modalities under the GKT paradigm, thereby supporting the study of GSR–overt modality relationships while also accommodating multimodal deception detection research.

2.2 Cross-Modal Knowledge Distillation

Knowledge distillation (KD) is an effective technique for transferring knowledge from one neural network to another [12]. Its core mechanism is a teacher–student framework, in which the student is trained to mimic the teacher by aligning teacher knowledge at different distillation locations, such as logits and intermediate features, under various objectives including divergence-based, regression-based, and relation-preserving losses [16, 31, 32, 39]. Cross-modal knowledge distillation (CMKD) extends KD to the setting where the teacher and student receive inputs from different modalities [14]. A long-standing challenge in CMKD is that the large discrepancy between modalities often makes conventional KD methods less effective when directly applied to heterogeneous teacher–student pairs [18]. To alleviate modality heterogeneity, existing CMKD methods have explored a variety of strategies. Some methods improve transferability by reshaping teacher knowledge into forms that are less modality-specific and easier for the student to learn, such as knowledge decomposition [18, 49], shared-knowledge distillation [26, 46], intermediate feature alignment [47], and relational or prototype-level distillation [24]. Other methods improve how knowledge is conveyed to the student by introducing better matched transfer mechanisms, such as bridging modules for easing cross-modal transfer [43], proxy or assistant models [18, 44], and adaptive teacher–student assignment strategies, including adaptive teacher selection [23] and dynamic teacher–student matching [41].

While these methods have demonstrated promising effectiveness, most of them still rely on predefined distillation schemes and do not explicitly model CMKD as a dynamically evolving process throughout training. In practice, however, the teacher–student gap is not static. As training progresses, it may change substantially, which in turn affects both what knowledge is more learnable and what transfer process is more suitable at different stages. Motivated by this observation, we propose a dynamic CMKD framework that explicitly models cross-modal distillation as an evolving process

during training. The framework adaptively regulates both transfer pathways and distillation signals according to the changing teacher–student representation gap.

3 MuDD Dataset

The Multimodal Deception Detection (MuDD) dataset was collected under the Guilty Knowledge Test (GKT) paradigm, which assumes that concealed information elicits stronger physiological responses to probe stimuli than to irrelevant stimuli. Accordingly, we formulate the task as concealed-number identification. In each trial, a participant privately selects one target number from 1 to 10. During testing, the ten candidate numbers are presented in randomized order over two rounds, each containing one probe and nine irrelevant numbers. All questions are delivered through pre-recorded professional interrogation audio to standardize stimulus presentation and reduce unintended interference [20], followed by a 10-second interval to capture the rise-and-recovery dynamics of the electrodermal response [11, 28]. Participants respond deceptively only to the selected number and truthfully to all others. Each participant contributes 20 dialogue clips, including 2 deceptive clips and 18 truthful clips, with both clip-level binary deception labels and trial-level 10-class concealed-number labels. The dataset includes synchronized visual, audio, and physiological recordings, together with Big-Five personality scores as subject-level trait annotations. Specifically, facial and upper-body behaviors are recorded at 30 fps, speech at 48 kHz, and physiological signals including GSR, PPG, and HR at 256 Hz. MuDD contains valid data from 130 native Chinese-speaking participants, totaling 690 minutes of synchronized multimodal recordings. All participants were recruited from a university student population. The study was approved by the institutional ethics review board, and written informed consent was obtained from all participants. The dataset is available for research purposes upon reasonable request and approval.²

4 Methodology

4.1 Problem formulation

For each subject, MuDD contains $Q = 20$ aligned question segments. We denote the student input sequence as $\mathcal{X} = \{x_q\}_{q=1}^Q$, where each x_q denotes the feature representation of the target non-contact modality for segment q , instantiated as either visual or audio features in our experiments, and the teacher GSR sequence as $\mathcal{G} = \{g_q\}_{q=1}^Q$. Each segment is labeled with $y_q \in \{0, 1\}$, where 1 indicates denial of the concealed item and 0 indicates denial of an irrelevant item. At the trial level, each subject has a concealed digit label $d^* \in \{1, \dots, 10\}$. Accordingly, our goal is to learn question-level deception detection from non-contact signals and, based on the aggregated deception evidence across the 20 questions, perform trial-level concealed-digit identification.

4.2 Overview of GPD

An overview of the proposed GPD framework is shown in Fig. 2. GPD consists of four components: a teacher network, a student network, a progressive knowledge distillation module, and a hierarchical task supervision module. The teacher takes GSR as input,

while the student takes the target non-contact modality as input, instantiated as either visual or audio features in our experiments. For each question segment, both networks produce feature representations and deception logits for cross-modal knowledge transfer. The progressive distillation module defines four candidate routes for transferring feature-level and logit-level teacher knowledge. During training, dynamic routing selects the route according to the teacher–student gap, and progressive weighting adjusts the relative importance of the two knowledge types. The hierarchical task supervision module jointly performs question-level and concealed-digit-level supervision in one framework. Binary deception labels supervise question-level predictions, while the two question-level deception scores for each digit are aggregated into digit-level evidence for concealed-digit supervision.

4.3 Teacher and student networks

The teacher and student networks share a similar prediction pipeline but operate on different modalities. For each segment q , the teacher network $\mathcal{T}$ takes the GSR input g_q , while the student network $\mathcal{S}$ takes the target non-contact input x_q , instantiated as either visual or audio features in our experiments. They produce two-dimensional logit vectors and intermediate feature representations as

$$(z_q^T, f_q^T) = \mathcal{T}(g_q), \quad (z_q^S, f_q^S) = \mathcal{S}(x_q), \quad (1)$$

where $z_q^T, z_q^S \in \mathbb{R}^2$ are the output logits used for logit-level knowledge distillation, $f_q^T \in \mathbb{R}^{D_T}$ and $f_q^S \in \mathbb{R}^{D_S}$ are the intermediate feature embeddings used for feature-level knowledge transfer. The teacher and student deceptive-class probabilities are defined as

$$t_q = \text{softmax}(z_q^T)_1, \quad p_q = \text{softmax}(z_q^S)_1, \quad (2)$$

where t_q and p_q denote the teacher and student probabilities of the deceptive class, respectively, and class label 1 denotes deception.

4.4 Progressive knowledge distillation

GPD distills two types of teacher knowledge: feature-level representations and logit-level predictions. Different scheduling patterns of these two knowledge types over the course of training define four route configurations, $\mathcal{R} = \{0, 1, 2, 3\}$, corresponding to no-feature, logit-first, joint, and feature-first distillation, respectively. Specifically, Route 0 disables feature-level distillation, Route 1 prioritizes logit-level distillation over feature-level distillation, Route 2 applies both jointly, and Route 3 prioritizes feature-level distillation over logit-level distillation. Dynamic routing selects the route configuration, and progressive weighting determines the relative contributions of the two knowledge types.

Gap-aware dynamic routing. Let $F_e^T = \{f_q^T\}_{q=1}^Q$ and $F_e^S = \{f_q^S\}_{q=1}^Q$ denote the teacher and student feature sets collected over the Q question segments at epoch e , respectively. The routing gate at epoch e is the teacher–student gap g_e , computed from the Centered Kernel Alignment (CKA) similarity between F_e^S and F_e^T as

$$\tilde{g}_e = 1 - \text{CKA}(F_e^S, F_e^T), \quad (3)$$

$$g_e = \mu g_{e-1} + (1 - \mu) \tilde{g}_e, \quad (4)$$

where $\tilde{g}_e$ denotes the raw teacher–student gap and $\mu \in [0, 1)$ is the EMA momentum. Given the gate signal g_e , the route state is updated by a hysteretic threshold rule. Intuitively, a larger gap

²<https://github.com/PANPANKK/GPD>

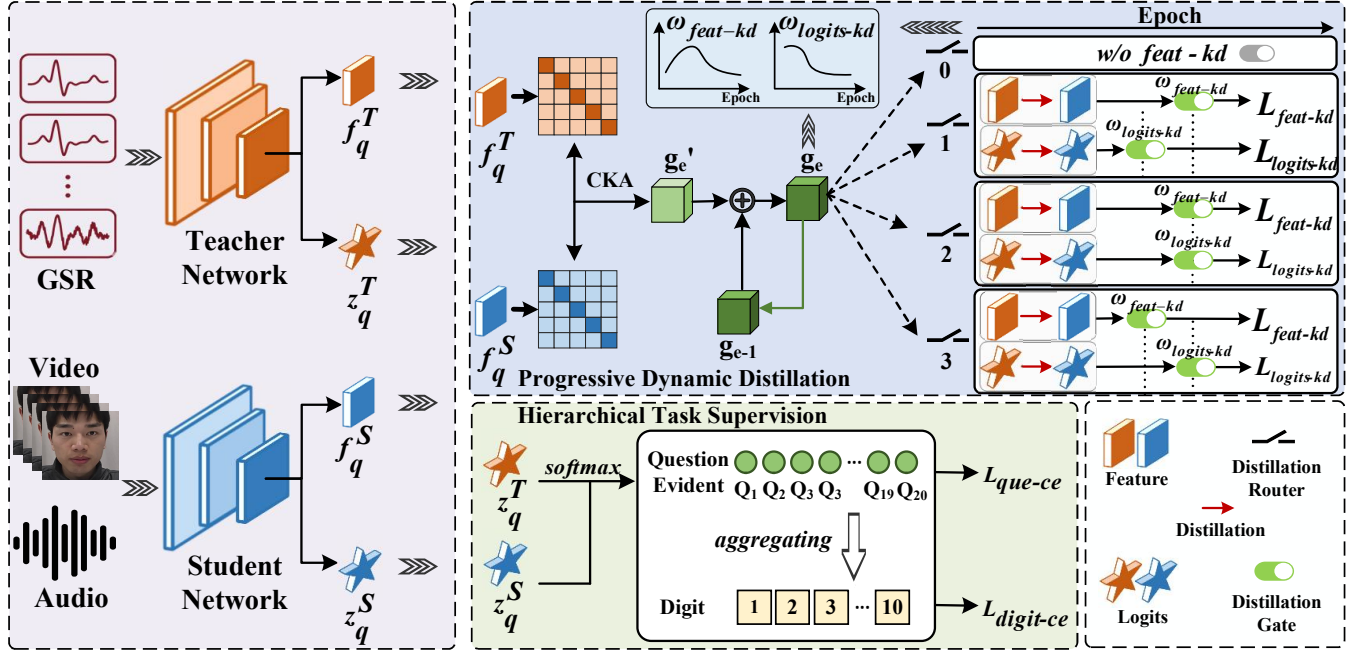


Figure 2: Overview of the proposed GSR-guided Progressive Distillation (GPD) framework.

favors more conservative distillation, while a smaller gap allows stronger feature-level transfer. Let $r_{e-1} \in \mathcal{R}$ denote the previous route state. We define the threshold set

$$\eta = (\eta_{nf}^{in}, \eta_{nf}^{out}, \eta_d^{in}, \eta_d^{out}, \eta_f^{out}, \eta_f^{in}), \quad (5)$$

with

$$\eta_{nf}^{in} \leq \eta_{nf}^{out} \leq \eta_d^{in} \leq \eta_d^{out} \leq \eta_f^{out} \leq \eta_f^{in}. \quad (6)$$

Let h_{e-1} be the current hold length and H the minimum hold length. If $h_{e-1} < H$, the route state remains unchanged, with $r_e = r_{e-1}$, to avoid excessively frequent route switching. Otherwise, the route is updated only when g_e crosses the corresponding entry threshold of a neighboring state; if g_e falls inside a hysteresis interval between the in/out thresholds, we retain the previous state, i.e., $r_e = r_{e-1}$. The threshold-based mapping is

$$r_e = \Phi(g_e; \eta) = \begin{cases} 3, & g_e \leq \eta_{nf}^{in}, \\ 2, & \eta_{nf}^{out} \leq g_e < \eta_d^{in}, \\ 1, & \eta_d^{out} \leq g_e < \eta_f^{out}, \\ 0, & g_e \geq \eta_f^{in}. \end{cases} \quad (7)$$

Progressive weighting. Given the selected route r_e , the logit-level and feature-level distillation weights are defined as

$$w_l(e) = \sigma\left(\frac{e - \beta_l(r_e)}{\tau_l}\right), \quad (8)$$

$$w_f(e) = \sigma\left(\frac{e - \beta_f(r_e)}{\tau_f}\right) \alpha_{\text{gap}}(g_e) \mathbb{I}[r_e \neq 0].$$

where $\sigma(\cdot)$ denotes the sigmoid function, $\beta_l(r_e)$ and $\beta_f(r_e)$ are route-dependent onset hyperparameters controlling when logit-level and feature-level distillation begin to take effect, respectively,

and τ_l, τ_f control the smoothness of the corresponding transitions. The feature-level weight is further modulated by a gap-aware factor

$$\alpha_{\text{gap}}(g_e) = \begin{cases} 1, & g_e \leq g_{\text{low}}, \\ \frac{g_{\text{high}} - g_e}{g_{\text{high}} - g_{\text{low}}}, & g_{\text{low}} < g_e < g_{\text{high}}, \\ 0, & g_e \geq g_{\text{high}}, \end{cases} \quad (9)$$

where g_{low} and g_{high} are the lower and upper gap thresholds, respectively, controlling the progressive activation of feature-level distillation from suppressed to fully enabled as the teacher-student gap decreases and the two representations become more compatible over the course of training in a stable manner.

Distillation objective. Under the selected route configuration, the feature-level distillation objective is defined as

$$\tilde{f}_q^S = g_\phi(f_q^S), \quad \mathcal{L}_{\text{feat-kd}} = \frac{1}{Q} \sum_{q=1}^Q \|\tilde{f}_q^S - f_q^T\|_2^2, \quad (10)$$

where g_ϕ is a learnable projection head, f_q^S is the student embedding, f_q^T is the teacher feature, and $\tilde{f}_q^S$ is the projected student feature in the teacher space. Logit-level knowledge distillation is defined by

$$\mathcal{L}_{\text{logit-kd}} = \tau_{\text{kd}}^2 \text{KL} \left(\text{softmax} \left(\frac{\mathbf{E}^T}{\tau_{\text{kd}}} \right) \parallel \text{softmax} \left(\frac{\mathbf{E}^S}{\tau_{\text{kd}}} \right) \right), \quad (11)$$

where $\mathbf{E}^T = [E_1^T, \dots, E_{10}^T]$ and $\mathbf{E}^S = [E_1^S, \dots, E_{10}^S]$ denote the teacher and student digit-level evidence vectors, respectively, in which E_d^T and E_d^S represent the aggregated evidence scores for digit d , obtained by summing question-level evidence across all questions querying that digit. The route-dependent distillation objective is

$$\mathcal{L}_{\text{kd}}(e) = \lambda_f w_f(e) \mathcal{L}_{\text{feat-kd}} + \lambda_l w_l(e) \mathcal{L}_{\text{logit-kd}}, \quad (12)$$

where λ_f and λ_l are static trade-off coefficients balancing them.

Hierarchical Task Supervision. The student is further supervised by ground-truth labels at both the question and digit levels:

$$\begin{aligned}\mathcal{L}_{\text{que-ce}} &= -\frac{1}{Q} \sum_{q=1}^Q \left[y_q \log p_q + (1 - y_q) \log(1 - p_q) \right], \\ \mathcal{L}_{\text{digit-ce}} &= -\sum_{d=1}^{10} \mathbb{I}[d = d^*] \log \frac{\exp(E_d^S)}{\sum_{j=1}^{10} \exp(E_j^S)},\end{aligned}\quad (13)$$

where $y_q \in \{0, 1\}$ is the question-level deception label, d^* is the ground-truth concealed digit, and $\mathbb{I}[\cdot]$ is the indicator function.

4.5 Overall objective and optimization

The overall training objective of GPD is defined as

$$\mathcal{L} = \mathcal{L}_{\text{que-ce}} + \lambda_d \mathcal{L}_{\text{digit-ce}} + \lambda_c \mathcal{L}_{\text{kd}}(e), \quad (14)$$

where λ_d and λ_c are trade-off coefficients. During training, the teacher is frozen, while the student network and the projection head g_ϕ are optimized by backpropagation.

5 Experiments

5.1 Experimental Setting

Implementation details. All experiments are conducted on a single NVIDIA GeForce RTX 4090 GPU. We train GPD for 120 epochs using Adam with a batch size of 16, an initial learning rate of 1×10^{-4} , and a weight decay of 1×10^{-4} . Unless otherwise specified, experiments are conducted with random seed 42. For the main comparison with state-of-the-art CMKD methods, all methods are repeated over five runs with different random seeds, and the results are reported as mean $\pm$ std. For the student model, the hidden dimension and re-embedding dimension are set to 512 and 256, respectively, and the dropout rate is set to 0.3. The trade-off coefficients are set to $\lambda_d = 0.3$, $\lambda_c = 1.0$, $\lambda_l = 0.7$, and $\lambda_f = 0.2$, and the distillation temperature is set to $\tau_{\text{kd}} = 2.0$. For gap-aware dynamic routing and progressive weighting, we set $\mu = 0.8$ and $H = 3$. The gap-aware weighting thresholds ($g_{\text{low}}, g_{\text{high}}$) are set to (0.40, 0.62) for video and (0.46, 0.76) for audio. The routing thresholds ($\eta_{nf}^{\text{in}}, \eta_{nf}^{\text{out}}, \eta_d^{\text{in}}, \eta_d^{\text{out}}, \eta_f^{\text{in}}, \eta_f^{\text{out}}$) are set to (0.34, 0.40, 0.40, 0.48, 0.52, 0.60) for video and (0.40, 0.46, 0.54, 0.60, 0.68, 0.76) for audio.

Preprocessing and feature extraction. For GSR, we apply a 2 Hz low-pass filter, downsample the signal from 256 Hz to 32 Hz, and resize each segment to 448 sampling points by truncation or zero-padding. The processed GSR is then fed into a teacher encoder adapted from [7], producing an 80-dimensional representation. For the visual modality, following GENLIE [48], we detect and align face regions based on facial landmarks, divide the aligned video into temporal segments, sample 16 frames per segment, and extract 768-dimensional visual features using VideoMAEv2. For the audio modality, frame-level acoustic features are extracted using pretrained WavLM [6], and local average pooling is applied according to the visual segment boundaries to obtain temporally aligned 768-dimensional audio features.

Evaluation protocol. We evaluate all methods under 5-fold cross-validation. Detailed fold assignments and per-fold statistics are

provided in the extended arXiv version of this paper. For question-level binary deception detection, we report F1 score and AUC, treating deception as the positive class. F1 measures the balance between precision and recall, while AUC measures class separability across decision thresholds. For trial-level concealed-digit identification, we report Top-1 and Top-2 accuracy, indicating whether the ground-truth concealed digit is ranked within the top one or top two predictions, respectively. Since the numbers of samples differ slightly across folds, final results are reported as sample-weighted averages over the five folds.

5.2 Main Results

Baseline Models. We compare GPD against nine representative state-of-the-art CMKD methods and the original non-distilled GENLIE baseline. To isolate the effect of different distillation strategies, all methods are implemented with the same student backbone and evaluated under the same 5-fold cross-validation protocol. GENLIE [48] serves as the non-distilled student backbone. The CMKD baselines include XKD [34], which performs self-supervised audio-visual distillation; SemBridge [25], which uses contrastive cross-modal distillation; HKD-Emotion [37], a hierarchical distillation framework; DecomKD [26], a decomposed distillation framework; CorrKD [24], a correlation-decoupled distillation framework; CDGKD [44], a densely guided distillation framework; C2KD [18], which adopts bidirectional cross-modal distillation; MST-Distill [23], a multi-stage teacher distillation framework; and LogitStd [36], which improves distillation via logit standardization. The comparison results are reported in Table 2.

As shown in Table 2, GPD achieves the best overall performance among all compared methods. In particular, its gains are more pronounced on digit-level prediction, where it achieves the highest Top-1 and Top-2 accuracy. This suggests that the proposed digit-level evidence transfer is effective in conveying the teacher’s concealed-digit discrimination ability to the non-contact student. At the same time, GPD also maintains the strongest F1/AUC results for question-level deception detection, indicating that the transferred knowledge remains beneficial at the finer-grained binary decision level. Another notable observation is that several existing CMKD baselines fail to consistently outperform the plain GENLIE model, and some even exhibit negative transfer. This suggests that under the large gap between physiological GSR signals and non-contact modalities, existing distillation strategies may be less effective because they typically rely on predefined transfer designs and do not explicitly model the evolving teacher–student discrepancy during training. In contrast, GPD models both what knowledge is transferred and how the transfer process is scheduled during training. The feature-level and logit-level objectives provide complementary supervision on intermediate representations and prediction-level signals, while dynamic routing and progressive weighting allow the transfer path and supervision strength to adapt to the evolving teacher–student discrepancy. Additional results, including the standalone performance of the GSR teacher, are provided in the extended arXiv version of this paper. It is also worth noting that the gains of GPD are not obtained by increasing model size.

Table 2: Main comparison with state-of-the-art CMKD methods on MuDD. Results are reported as mean $\pm$ std over five runs with different random seeds. The best results are shown in bold, and the second-best results are underlined.

Method	Params (M)	FLOPs/sample (M)	Video modality				Audio modality			
			Top-1	Top-2	F1	AUC	Top-1	Top-2	F1	AUC
GENLIE	0.908	36.75	<u>26.8$\pm$5.0</u>	35.3 $\pm$ 5.7	<u>16.2$\pm$2.8</u>	<u>54.4$\pm$1.4</u>	29.9 $\pm$ 6.7	39.8 $\pm$ 5.1	18.1 $\pm$ 4.8	56.1 $\pm$ 4.2
XKD	1.237	49.32	23.6 $\pm$ 6.5	<u>38.4$\pm$4.5</u>	13.9 $\pm$ 4.9	53.3 $\pm$ 1.9	28.9 $\pm$ 9.7	39.1 $\pm$ 8.6	15.3 $\pm$ 4.2	55.1 $\pm$ 6.4
SemBridge	1.456	34.16	23.7 $\pm$ 5.7	32.9 $\pm$ 5.1	12.9 $\pm$ 7.1	52.5 $\pm$ 3.6	28.4 $\pm$ 6.7	<u>39.9$\pm$6.3</u>	<u>18.6$\pm$5.2</u>	<u>58.3$\pm$2.9</u>
HKD-Emotion	1.075	34.16	24.4 $\pm$ 7.3	33.7 $\pm$ 5.1	8.1 $\pm$ 6.9	52.3 $\pm$ 1.8	29.9 $\pm$ 9.0	36.1 $\pm$ 11.5	7.6 $\pm$ 7.8	58.0 $\pm$ 2.0
DecomKD	1.535	61.78	24.5 $\pm$ 6.0	32.1 $\pm$ 8.8	5.8 $\pm$ 5.7	52.8 $\pm$ 2.5	<u>30.0$\pm$7.2</u>	39.3 $\pm$ 8.1	14.5 $\pm$ 7.6	58.5 $\pm$ 3.8
CorrKD	1.075	42.40	22.3 $\pm$ 2.3	33.7 $\pm$ 4.7	11.0 $\pm$ 5.1	52.6 $\pm$ 2.0	27.6 $\pm$ 5.6	38.4 $\pm$ 6.7	18.3 $\pm$ 3.3	<u>58.8$\pm$2.9</u>
CDGKD	1.209	48.22	24.5 $\pm$ 5.8	37.6 $\pm$ 6.8	13.7 $\pm$ 5.9	53.5 $\pm$ 2.0	27.6 $\pm$ 6.6	39.1 $\pm$ 7.7	15.1 $\pm$ 8.2	56.7 $\pm$ 2.4
C2KD	0.944	37.69	23.8 $\pm$ 2.3	33.1 $\pm$ 5.6	12.7 $\pm$ 3.2	52.4 $\pm$ 3.1	<u>30.0$\pm$4.3</u>	36.9 $\pm$ 6.7	18.4 $\pm$ 3.5	57.7 $\pm$ 3.2
MST-Distill	1.049	34.16	23.1 $\pm$ 2.3	34.6 $\pm$ 3.2	8.5 $\pm$ 4.9	51.1 $\pm$ 3.1	28.5 $\pm$ 4.4	37.6 $\pm$ 7.6	17.5 $\pm$ 4.9	57.6 $\pm$ 4.2
LogitStd	<u>0.921</u>	<u>36.32</u>	24.6 $\pm$ 4.4	30.8 $\pm$ 6.3	5.5 $\pm$ 6.0	52.6 $\pm$ 3.1	28.4 $\pm$ 6.7	<u>39.9$\pm$6.3</u>	18.5 $\pm$ 5.2	57.3 $\pm$ 2.9
GPD (Ours)	0.908	36.79	32.3$\pm$4.4	40.1$\pm$4.5	16.7$\pm$3.8	54.6$\pm$2.3	35.3$\pm$4.9	43.8$\pm$6.6	20.1$\pm$2.3	59.9$\pm$1.85

Table 3: Ablation study of different components in GPD on MuDD. Prog. wt. denotes progressive weighting.

Method	Video modality				Audio modality			
	Top-1	Top-2	F1	AUC	Top-1	Top-2	F1	AUC
GENLIE	26.92	35.38	16.30	53.14	29.86	40.77	18.94	59.03
w/o $\mathcal{L}_{\text{logit-kd}}$	26.92	37.69	17.00	52.13	30.77	41.27	18.50	57.48
w/o $\mathcal{L}_{\text{feat-kd}}$	25.38	36.92	16.70	51.46	32.31	43.08	19.70	57.66
w/o prog. wt.	23.08	35.38	16.72	51.89	30.77	42.47	20.05	59.25
GPD	32.31	40.00	17.43	53.55	35.38	43.84	20.63	59.30

5.3 Component and Strategy Ablation

Q1: Are the two distillation objectives and progressive weighting necessary? We first examine the contribution of each core component in GPD. Starting from the original student baseline GENLIE, we compare the full model against three variants: w/o $\mathcal{L}_{\text{logit-kd}}$, w/o $\mathcal{L}_{\text{feat-kd}}$, and w/o *progressive weighting*. As shown in Table 3, GPD achieves the best overall performance in both the video and audio modalities. Removing $\mathcal{L}_{\text{logit-kd}}$ or $\mathcal{L}_{\text{feat-kd}}$ consistently degrades the results, indicating that the two distillation objectives provide complementary supervision. Removing progressive weighting also leads to inferior performance, confirming its importance for stable knowledge transfer across the large teacher–student modality gap.

Q2: Which progressive weighting strategy is most effective? We compare four progressive weighting strategies while keeping the architecture and the two distillation objectives unchanged: sigmoid, which increases the weights smoothly with a slow start and saturation; linear, which increases them at a constant rate; step, which activates them abruptly at predefined stages; and cosine, which follows a smooth nonlinear schedule. As shown in Table 4, the sigmoid schedule achieves the best overall performance. We attribute this to its smooth transition, which is better suited to coordinating the two complementary distillation objectives. Compared with linear, step, and cosine schedules, sigmoid weighting provides a more stable balance between $\mathcal{L}_{\text{feat-kd}}$ and $\mathcal{L}_{\text{logit-kd}}$ throughout training.

Table 4: Comparison of different progressive weighting schedules on MuDD.

Schedule	Video modality				Audio modality			
	Top-1	Top-2	F1	AUC	Top-1	Top-2	F1	AUC
Linear	30.77	41.54	17.27	52.67	33.08	38.46	18.97	57.07
Step	30.00	41.54	17.74	52.80	34.62	40.76	19.22	57.41
Cosine	30.00	39.23	17.53	52.70	33.08	40.00	19.06	58.20
Sigmoid	32.31	40.00	17.43	53.55	35.38	43.84	20.63	59.30

Q3: Is dynamic routing better than fixed route configurations? We compare the full model with dynamic routing (Dyn.) against four fixed route configurations: no-feature (No-feat.), logit-first, joint, and feature-first (Feat.-first). Table 5 shows that dynamic routing achieves the best overall performance. We attribute this to the fact that the teacher–student gap evolves during training, making a single fixed route suboptimal across stages. When the gap is large, more conservative routes are preferable, whereas smaller gaps allow stronger feature-level transfer. Dynamic routing adapts the transfer path to this changing optimization state, while fixed configurations impose the same distillation order throughout training. This interpretation is supported by the behavior of individual fixed routes: the *joint* route may introduce conflicting supervision when the teacher–student gap is still large, while *feature-first* can be too aggressive and *logit-first* too conservative for effective transfer throughout training. Although *no-feature* remains competitive on some video metrics, its lack of feature-level guidance limits overall performance, especially for audio.

5.4 Interaction Between Routing and Weighting

Q4: Are dynamic routing and progressive weighting complementary? To decouple the effects of routing and progressive weighting, we toggle the routing mechanism on and off under different weighting schedules. As shown in Table 6, dynamic routing works best when combined with sigmoid weighting. A possible reason is that

Table 5: Comparison of different distillation routes on MuDD.

Route	Video modality				Audio modality			
	Top-1	Top-2	F1	AUC	Top-1	Top-2	F1	AUC
No-feat.	30.77	41.54	17.68	52.69	34.62	41.54	19.75	57.47
Feat.-first	31.31	39.23	17.31	52.81	33.08	42.31	19.42	58.14
Logit-first	31.54	40.77	17.47	52.46	33.08	39.23	19.14	56.73
Joint	30.77	39.31	17.46	52.66	32.31	42.31	17.92	58.13
Dyn.	32.31	40.00	17.43	53.55	35.38	43.84	20.63	59.30

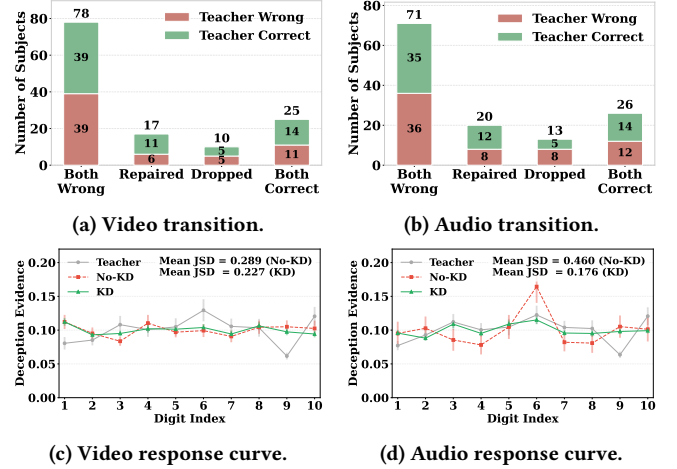
Table 6: Decoupled ablation of routing and weighting on MuDD. “Wt.” and “Rt.” denote weighting schedule and route configuration, while “On” and “Off” indicate whether dynamic routing is enabled. For each weighting schedule, the better result between On and Off is highlighted in bold.

Wt.	Rt.	Video modality				Audio modality			
		Top-1	Top-2	F1	AUC	Top-1	Top-2	F1	AUC
Sig.	On	32.31	40.00	17.43	53.55	35.38	43.84	20.63	59.30
	Off	30.77	42.31	17.37	52.66	33.08	42.31	19.42	58.14
Lin.	On	30.77	41.54	17.27	52.67	33.08	38.46	18.97	57.07
	Off	31.54	40.77	17.26	52.52	31.54	41.54	17.77	56.68
Step	On	30.00	41.54	17.74	52.80	34.62	40.76	19.22	57.41
	Off	30.77	40.77	17.94	52.77	32.30	40.00	20.02	57.68
Cos.	On	30.00	39.23	17.53	52.70	33.08	40.00	19.06	58.20
	Off	30.77	40.00	17.21	52.59	32.31	40.77	18.71	58.02

the two strategies regulate different aspects of the transfer process: routing determines which route configuration is used at a given stage, whereas weighting controls how strongly the selected route influences optimization. Under linear, step, and cosine schedules, the benefit of routing becomes less consistent, suggesting weaker coordination between route selection and distillation strength. Overall, these results indicate that effective transfer depends not only on selecting an appropriate route configuration, but also on pairing it with a weighting strategy that supports stable optimization.

5.5 Effect of Distillation on Student Behavior

Q5: How does distillation affect the student’s decision behavior? We analyze both video and audio students using prediction-transition statistics and representative response curves, as shown in Fig. 3. The transition plots summarize prediction changes before and after distillation, including *both-wrong*, *repaired*, *dropped*, and *both-correct* cases, each further divided according to whether the teacher prediction is correct. The response curves show the distribution of digit-level evidence over the 10 candidate digits within each trial, reflecting how the student organizes trial-level evidence before and after distillation. As shown in Fig. 3, repaired samples outnumber dropped ones in both modalities, indicating that distillation improves the student more often than it harms it. Moreover, most repaired cases are teacher-consistent (11/17 for video and 12/20 for

**Figure 3: Effect of distillation on student behavior in the video and audio modalities.**

audio), suggesting that the gains mainly come from effective teacher guidance rather than incidental prediction changes. Some dropped cases are also teacher-consistent, indicating that distillation can occasionally pull the student toward the teacher even when this does not improve the ground-truth decision. For audio, dropped cases contain a larger proportion of teacher-wrong samples (8/13), suggesting higher sensitivity to local teacher noise. The response curves further show that the student after distillation aligns more closely with the teacher, especially around salient peaks and in the overall response trend. This suggests that distillation improves not only final predictions but also the student’s trial-level evidence organization, making its responses more consistent with the physiological teacher over trials.

6 Conclusion

We addressed non-contact deception detection from the perspective of physiological-signal-guided cross-modal distillation. Specifically, we introduced MuDD, a large-scale multimodal deception detection dataset collected under the Guilty Knowledge Test paradigm, and proposed GPD, a GSR-guided progressive distillation framework that uses gap-aware dynamic routing and progressive weighting to adapt both the transfer pathway and the transfer strength according to the evolving teacher–student gap. Extensive experiments show that GPD effectively performs joint modeling of the two tasks and consistently outperforms representative cross-modal knowledge distillation (CMKD) baselines. These results indicate that GSR contains transferable and highly informative deception-related knowledge, and that modeling cross-modal distillation as a dynamic rather than predefined process enables more effective knowledge transfer under substantial modality discrepancy.

A limitation of the current work is that it does not explicitly model the alignment between physiological responses in GSR and overt deception cues in non-contact modalities. These cross-modal relations are currently captured implicitly through knowledge transfer. Future work will explore explicit alignment modeling to improve both interpretability and detection performance.

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References

- [1] Triantafyllos Afouras, Joon Son Chung, and Andrew Zisserman. 2020. Asr is all you need: Cross-modal distillation for lip reading. In *ICASSP 2020-2020 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*. IEEE, Piscataway, NJ, USA, 2143–2147.
- [2] Muhammad Haseeb Aslam, Muhammad Osama Zeeshan, Soufiane Belharbi, Marco Pedersoli, Alessandro Lameiras Koerich, Simon Bacon, and Eric Granger. 2024. Distilling privileged multimodal information for expression recognition using optimal transport. In *2024 IEEE 18th International Conference on Automatic Face and Gesture Recognition (FG)* (Istanbul, Turkey). IEEE, Piscataway, NJ, USA, 1–10. doi:10.1109/FG59268.2024.10582047
- [3] Gershon Ben-Shakhar and Eitan Elaad. 2003. The validity of psychophysiological detection of information with the Guilty Knowledge Test: A meta-analytic review. *Journal of Applied Psychology* 88, 1 (2003), 131.
- [4] Charles F Bond Jr and Bella M DePaulo. 2006. Accuracy of deception judgments. *Personality and social psychology Review* 10, 3 (2006), 214–234.
- [5] Cong Cai, Shan Liang, Xuefei Liu, Kang Zhu, Zhengqi Wen, Jianhua Tao, Heng Xie, Jizhou Cui, Yiming Ma, Zhenhua Cheng, Hanzhe Xu, Ruibo Fu, Bin Liu, and Yongwei Li. 2025. MDPE: A Multimodal Deception Dataset with Personality and Emotional Characteristics. In *Proceedings of the 33rd ACM International Conference on Multimedia* (Dublin, Ireland) (MM '25). Association for Computing Machinery, New York, NY, USA, 12957–12964. doi:10.1145/3746027.3758242
- [6] Sanyuan Chen, Chengyi Wang, Zhengyang Chen, Yu Wu, Shujie Liu, Zhuo Chen, Jinyu Li, Naoyuki Kanda, Takuya Yoshioka, Xiong Xiao, et al. 2022. Wavlm: Large-scale self-supervised pre-training for full stack speech processing. *IEEE Journal of Selected Topics in Signal Processing* 16, 6 (2022), 1505–1518.
- [7] Ziqiang Cheng, Yang Yang, Shuo Jiang, Wenjie Hu, Zhangchi Ying, Ziwei Chai, and Chunping Wang. 2021. Time2Graph+: Bridging time series and graph representation learning via multiple attentions. *IEEE Transactions on Knowledge and Data Engineering* 35, 2 (2021), 2078–2090.
- [8] Bella M DePaulo, Deborah A Kashy, Susan E Kirkendol, Melissa M Wyer, and Jennifer A Epstein. 1996. Lying in everyday life. *Journal of personality and social psychology* 70, 5 (1996), 979.
- [9] Mingyu Ding, An Zhao, Zhiwu Lu, Tao Xiang, and Ji-Rong Wen. 2019. Face-focused cross-stream network for deception detection in videos. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*. IEEE, Piscataway, NJ, USA, 7802–7811.
- [10] Laslo Dinges, Marc-André Fiedler, Ayoub Al-Hamadi, Thorsten Hempel, Ahmed Abdelrahman, Joachim Weimann, Dmitri Bershadsky, and Johann Steiner. 2024. Exploring facial cues: automated deception detection using artificial intelligence. *Neural Computing and Applications* 36, 24 (2024), 14857–14883.
- [11] Don C Fowles, Margaret J Christie, Robert Edelberg, William W Grings, David T Lykken, and Peter H Venables. 1981. Publication recommendations for electrodermal measurements. *Psychophysiology* 18, 3 (1981), 232–239.
- [12] Jianping Gou, Baosheng Yu, Stephen J Maybank, and Dacheng Tao. 2021. Knowledge distillation: A survey. *International journal of computer vision* 129, 6 (2021), 1789–1819.
- [13] Xiaobao Guo, Nithish Muthuchamy Selvaraj, Zitong Yu, Adams Wai-Kin Kong, Bingquan Shen, and Alex Kot. 2023. Audio-visual deception detection: Dols dataset and parameter-efficient crossmodal learning. In *Proceedings of the IEEE/CVF International Conference on Computer Vision*. IEEE, Piscataway, NJ, USA, 22135–22145.
- [14] Saurabh Gupta, Judy Hoffman, and Jitendra Malik. 2016. Cross modal distillation for supervision transfer. In *Proceedings of the IEEE conference on computer vision and pattern recognition*. IEEE, Piscataway, NJ, USA, 2827–2836.
- [15] Viresh Gupta, Mohit Agarwal, Manik Arora, Tanmoy Chakraborty, Richa Singh, and Mayank Vatsa. 2019. Bag-of-lies: A multimodal dataset for deception detection. In *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition workshops*. IEEE, Piscataway, NJ, USA, 0–0.
- [16] Geoffrey Hinton, Oriol Vinyals, and Jeff Dean. 2015. Distilling the Knowledge in a Neural Network. arXiv:1503.02531. arXiv:1503.02531
- [17] Julia Hirschberg, Stefan Bensus, Jason M Brenier, Frank Enos, Sarah Friedman, Sarah Gilman, Cynthia Girand, Martin Graciarena, Andreas Kathol, Laura Michaelis, et al. 2005. Distinguishing deceptive from non-deceptive speech. In *Proc. Interspeech 2005*. International Speech Communication Association, Lisbon, Portugal, 1833–1836.
- [18] Fushuo Huo, Wenchao Xu, Jingcai Guo, Haozhao Wang, and Song Guo. 2024. C2kd: Bridging the modality gap for cross-modal knowledge distillation. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*. IEEE, Piscataway, NJ, USA, 16006–16015.
- [19] Gargi Joshi, Vaibhav Tasgaonkar, Aditya Deshpande, Aditya Desai, Bhavya Shah, Akshay Kushawaha, Aadith Sukumar, Kermi Kotecha, Saumit Kunder, Yoginii Waykole, et al. 2025. Multimodal machine learning for deception detection using behavioral and physiological data. *Scientific Reports* 15, 1 (2025), 8943.
- [20] FK Lahri and AK Ganguly. 1978. An experimental study of the accuracy of polygraph technique in diagnosis of deception with volunteer and criminal subjects. *Polygraph* 7 (1978), 89–94.
- [21] Sarah Ita Levitan, Guozhen An, Mandi Wang, Gideon Mendels, Julia Hirschberg, Michelle Levine, and Andrew Rosenberg. 2015. Cross-Cultural Production and Detection of Deception from Speech. In *Proceedings of the 2015 ACM Workshop on Multimodal Deception Detection* (Seattle, Washington, USA). Association for Computing Machinery, New York, NY, USA, 1–8.
- [22] Sarah Ita Levitan, Angel Maredia, and Julia Hirschberg. 2018. Linguistic cues to deception and perceived deception in interview dialogues. In *Proceedings of the 2018 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies, Volume 1 (Long Papers)*. Association for Computational Linguistics, New Orleans, Louisiana, 1941–1950.
- [23] Hui Li, Pengfei Yang, Juanyang Chen, Le Dong, Yanxin Chen, and Quan Wang. 2025. Mst-distill: Mixture of specialized teachers for cross-modal knowledge distillation. In *Proceedings of the 33rd ACM International Conference on Multimedia*. Association for Computing Machinery, New York, NY, USA, 1588–1597.
- [24] Mingcheng Li, Dingkan Yang, Xiao Zhao, Shuaibing Wang, Yan Wang, Kun Yang, Mingyang Sun, Dongliang Kou, Ziyun Qian, and Lihua Zhang. 2024. Correlation-decoupled knowledge distillation for multimodal sentiment analysis with incomplete modalities. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*. IEEE, Piscataway, NJ, USA, 12458–12468.
- [25] Hangyu Lin, Chen Liu, Chengming Xu, Zhengqi Gao, Yanwei Fu, and Yuan Yao. 2024. A Generalization Theory of Cross-Modality Distillation with Contrastive Learning. arXiv:2405.03355. arXiv:2405.03355 [cs.LG]
- [26] Yanfeng Liu and Lefei Zhang. 2025. Multimodal decomposed distillation with instance alignment and uncertainty compensation for thermal object detection. In *Proceedings of the 33rd ACM International Conference on Multimedia*. Association for Computing Machinery, New York, NY, USA, 2294–2303.
- [27] E Paige Lloyd, Jason C Deska, Kurt Hugenberg, Allen R McConnell, Brandon T Humphrey, and Jonathan W Kunstman. 2019. Miami University deception detection database. *Behavior research methods* 51, 1 (2019), 429–439.
- [28] David T Lykken. 1959. The GSR in the detection of guilt. *Journal of Applied Psychology* 43, 6 (1959), 385.
- [29] Merylin Monaro, Pasquale Capuozzo, Federica Ragucci, Antonio Maffei, Antonietta Curci, Cristina Scarpazza, Alessandro Angrilli, and Giuseppe Sartori. 2020. Using blink rate to detect deception: A study to validate an automatic blink detector and a new dataset of videos from liars and truth-tellers. In *International Conference on Human-Computer Interaction*. Springer, Cham, Switzerland, 494–509.
- [30] Jan Ondras and Hatice Gunes. 2018. Detecting deception and suspicion in dyadic game interactions. In *Proceedings of the 20th ACM international conference on multimodal interaction*. Association for Computing Machinery, New York, NY, USA, 200–209.
- [31] Wonpyo Park, Dongju Kim, Yan Lu, and Minsu Cho. 2019. Relational knowledge distillation. In *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*. IEEE, Piscataway, NJ, USA, 3967–3976.
- [32] Baoyun Peng, Xiao Jin, Jiaheng Liu, Dongsheng Li, Yichao Wu, Yu Liu, Shunfeng Zhou, and Zhaoqing Zhang. 2019. Correlation congruence for knowledge distillation. In *Proceedings of the IEEE/CVF international conference on computer vision*. IEEE, Piscataway, NJ, USA, 5007–5016.
- [33] Verónica Pérez-Rosas, Mohamed Abouelenen, Rada Mihalcea, and Mihai Burzo. 2015. Deception detection using real-life trial data. In *Proceedings of the 2015 ACM on international conference on multimodal interaction*. Association for Computing Machinery, New York, NY, USA, 59–66.
- [34] Pritam Sarkar and Ali Etamad. 2024. Xkd: Cross-modal knowledge distillation with domain alignment for video representation learning. In *Proceedings of the AAAI Conference on Artificial Intelligence*, Vol. 38. AAAI Press, Washington, DC, USA, 14875–14885.
- [35] Felix Soldner, Verónica Pérez-Rosas, and Rada Mihalcea. 2019. Box of Lies: Multimodal Deception Detection in Dialogues. In *Proceedings of the 2019 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies, Volume 1 (Long and Short Papers)*, Jill Burstein, Christy Doran, and Tamar Solorio (Eds.). Association for Computational Linguistics, Minneapolis, Minnesota, 1768–1777. doi:10.18653/v1/N19-1175
- [36] Shangquan Sun, Wenqi Ren, Jingzhi Li, Rui Wang, and Xiaochun Cao. 2024. Logit standardization in knowledge distillation. In *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*. IEEE, Piscataway, NJ, USA, 15731–15740.
- [37] Teng Sun, Yinwei Wei, Juntong Ni, Zixin Liu, Xuemeng Song, Yaowei Wang, and Liqiang Nie. 2024. Multi-modal emotion recognition via hierarchical knowledge distillation. *IEEE Transactions on Multimedia* 26 (2024), 9036–9046.

- [38] John Synnott, David Dietzel, and Maria Ioannou. 2015. A review of the polygraph: history, methodology and current status. *Crime Psychology Review* 1, 1 (2015), 59–83. arXiv:<https://doi.org/10.1080/23744006.2015.1060080> doi:10.1080/23744006.2015.1060080
- [39] Frederick Tung and Greg Mori. 2019. Similarity-preserving knowledge distillation. In *Proceedings of the IEEE/CVF international conference on computer vision*. IEEE, Piscataway, NJ, USA, 1365–1374.
- [40] Martina Vicianova. 2015. Historical techniques of lie detection. *Europe's journal of psychology* 11, 3 (2015), 522.
- [41] Hu Wang, Congbo Ma, Jianpeng Zhang, Yuan Zhang, Jodie Avery, Louise Hull, and Gustavo Carneiro. 2023. Learnable cross-modal knowledge distillation for multimodal learning with missing modality. In *International Conference on Medical Image Computing and Computer-Assisted Intervention*. Springer, Cham, Switzerland, 216–226.
- [42] Lin Wang and Kuk-Jin Yoon. 2021. Knowledge distillation and student-teacher learning for visual intelligence: A review and new outlooks. *IEEE transactions on pattern analysis and machine intelligence* 44, 6 (2021), 3048–3068.
- [43] Riling Wei, Kelu Yao, Chuanguang Yang, Jin Wang, Zhuoyan Gao, and Chao Li. 2025. Asymmetric Cross-Modal Knowledge Distillation: Bridging Modalities with Weak Semantic Consistency. arXiv:2511.08901. arXiv:2511.08901 doi:10.48550/arXiv.2511.08901
- [44] Shuang Wu, Heng Liang, Yong Zhang, Yanlin Chen, and Ziyu Jia. 2025. A cross-modal densely guided knowledge distillation based on modality rebalancing strategy for enhanced unimodal emotion recognition. In *Proceedings of the Thirty-Fourth International Joint Conference on Artificial Intelligence, IJCAI 2025, Montreal, Canada, August 16–22, 2025*. International Joint Conferences on Artificial Intelligence Organization, Montreal, Canada, 4236–4244.
- [45] Xiaolin Xu, Wenming Zheng, Hailun Lian, Sunan Li, Jiateng Liu, Anbang Liu, Cheng Lu, Yuan Zong, and Zongbao Liang. 2025. Multimodal lie detection dataset based on Chinese dialogue. *Journal of Image and Graphics* 30, 8 (2025), 2729–2742. doi:10.11834/jig.240571
- [46] Zihui Xue, Zhengqi Gao, Sucheng Ren, and Hang Zhao. 2023. The Modality Focusing Hypothesis: Towards Understanding Crossmodal Knowledge Distillation. In *The Eleventh International Conference on Learning Representations (ICLR 2023)*. OpenReview.net, Kigali, Rwanda, 21 pages.
- [47] Su Zhang, Chuangao Tang, and Cuntai Guan. 2022. Visual-to-EEG cross-modal knowledge distillation for continuous emotion recognition. *Pattern Recognition* 130 (2022), 108833. doi:10.1016/j.patcog.2022.108833
- [48] Zongshun Zhang, Yao Liu, Qiao Liu, Xuefeng Peng, Peiyuan Jiang, Jiaye Yang, Daibing Yao, and Wei Lin. 2026. GenLie: A Global-Enhanced Lie Detection Network under Sparsity and Semantic Interference. arXiv:2603.16935 [cs.CV] <https://arxiv.org/abs/2603.16935>
- [49] Borui Zhao, Quan Cui, Renjie Song, Yiyu Qiu, and Jiajun Liang. 2022. Decoupled knowledge distillation. In *Proceedings of the IEEE/CVF Conference on computer vision and pattern recognition*. IEEE, Piscataway, NJ, USA, 11953–11962.